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Psychological and Physiological Acoustics: Paper 1aPP11**Enhancing emotional well-being through active music listening: A study on mood improvement effects of music rhythm games**

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Anxiety and depression are growing problems worldwide, and the massive disruption of mental health care services during the COVID-19 pandemic has highlighted the need for more accessible and convenient digital tools for mental health care. Music therapy has been recognized as an effective non-invasive treatment for mental illnesses, but little research has been done on the therapeutic effects of active music therapy in digital formats. This paper investigates the effectiveness of active music therapy in a music rhythm game format for improving mood and examines the impact of five different song sequencing methods (random, all-relaxed, all-upbeat, upbeat-to-relaxed, and relaxed-to-upbeat) on therapeutic outcomes. Listening tests were conducted with 89 subjects (all university students in Hong Kong) to analyze changes in valence and arousal. The results indicate that all proposed song sequencing methods positively increased arousal levels, while all methods, except all-upbeat, significantly increased valence levels. All-relaxed and all-upbeat gave the best increase in arousal. Upbeat-to-relaxed and relaxed-to-upbeat gave the best increase in valence. All-relaxed and upbeat-to-relaxed were most effective in increasing both arousal and valence. The study also found that tailored recommendations based on subjects' initial arousal and valence levels and levels of anxiety and depression can further improve therapeutic outcomes.

1. INTRODUCTION

Anxiety and depression are common mental disorders worldwide and have raised growing concerns in recent years, especially since the emergence of COVID-19. According to statistics from the World Health Organization, the global prevalence of anxiety and depression has dramatically increased by 25% since the pandemic, and over 60% of 130 countries reported in a WHO survey that their mental health services have been severely interrupted, creating a huge gap in mental health care [1, 2]. The resulting inability to access face-to-face services has driven many people to turn to online support, signaling an urgent need for stable, easily accessible and effective digital alternatives to ensure people can receive appropriate mental support under difficult circumstances hereafter.

Music is often described as an “universal language of humankind” [3]. The intrinsic power of music in triggering and moving emotions has been proved repeatedly by numerous studies [4-8]. Music therapy is a powerful non-invasive means for mental therapeutic treatment. Unlike medication, music therapy does not require the intake of any chemical substances and thus will not adversely affect patients’ physical health. In addition to its mild nature, the effectiveness of music therapy in combating mental health issues is also well-supported by previous studies [9-13]. While traditional music therapy often requires the involvement of professional music therapists to guide musical activities, the service has become more inaccessible due to reduced face-to-face therapy sessions during the pandemic. Despite the fact that there are a number of music meditation applications on the market for relaxation and mood regulation, they usually only involve passive listening. Compared with simple passive listening, active listening where the user actively responds to the music in some way can bring even better therapeutic effects [14, 15]. It is worthwhile to explore effective approaches that provide active engagement in musical activity, and at the same time allow users to conveniently access them for self-help purposes.

Among all the digital formats, games often offer the most varied forms of engagement. Although many games employ music to evoke moods and build up emotions, rhythm games are the most closely tied to music of all game genres, and they offer the greatest opportunities for players to interact with the music through gameplay. Above all, music rhythm games are widely accepted [16] and easily deliverable due to their popularity and variety, making them a strong candidate as a modern therapeutic aid.

This paper aims to investigate whether active music therapy in a digital music rhythm game format is effective at improving mood, and if so, what types of song sequencing provide the best therapeutic effects. In this thesis, five song sequencing methods are proposed for sequencing songs in the music rhythm game: random, all-relaxed, all-upbeat, relaxed-to-upbeat, upbeat-to-relaxed. Listening tests were conducted to analyse the respective changes in valence and arousal levels in players before and after the gameplay as well as subjective perception of mood improvement of participants using different song sequencing methods.

2. RELATED WORK

A. WORK ON MUSIC THERAPY

Extensive research has been conducted on the effectiveness of music therapy in treating mental illnesses. There are various forms of music therapy, which can be mainly classified into two categories: passive and active listening. In passive listening, the client listens to music and responds to the experience afterwards either silently, verbally, or via another modality [17]. In active listening, the client usually engages in some kind of musical activities, for example repeating simple music rhythms or melodies, creating music with instruments under guidance, or composing songs [18-20].

Several studies have investigated the effectiveness of passive and active listening in various contexts, such as alleviating symptoms of Alzheimer’s disease [21], improving well-being and managing stress post-COVID-19 [22], and relieving pain and stress during medical treatment [23]. These studies have demonstrated the efficacy of passive and active music therapy methods. However, most of the previous studies on music therapy have been conducted in traditional face-to-face formats, with little use of modern technology to enhance the therapeutic process. As technology continues to advance, the combination of digital tools and music therapy has the potential to improve the effectiveness and efficiency of treating psychological issues, especially in situations where face-to-face services are disrupted or there is a shortage of professionals, as has been seen during the COVID-19 pandemic. Recognizing this gap, our group has taken the initiative to explore the use of digital tools in music therapy and has developed a prototype music therapy application to investigate the impact of passive listening on listeners’ valence and arousal levels using four different song sequencing methods: random, consoling, uplifting, and relaxing [24]. The results of our study support the therapeutic power of music therapy in a digital form, especially the effectiveness of the uplifting and relaxing methods in positively shifting listeners’ valence levels. This study highlights the potential of exploring more active digital music therapy tools such as music rhythm games and identifying effective song sequencing methods for enhancing the mood of listeners.

Intriguingly, some studies have compared passive and active listening techniques and have suggested that active listening may have greater overall impact by motivating participants to engage in the activity for longer durations and enhancing feelings of enjoyment and comfort [14, 15, 25]. However, most active listening activities involve playing musical instruments or composing songs, which can be challenging for subjects with little or no musical background and may prevent them from fully participating in active music therapy without expert supervision. It is therefore important to explore whether active music therapy can retain its therapeutic power in a digital format that requires minimal or no professional supervision, and if so, what kind of song sequencing methods are more effective at improving mood.

B. WORK ON GAME THERAPY AND MUSIC RHYTHM GAMES

In traditional active music therapy, activities include lyrics analysis, song writing, music games, facilitated instrument playing, singalong and more. Among these, music games require the least amount of musical background and guidance, making them an ideal starting point for exploring active music therapy. The use of digital games as a therapeutic tool is not a new concept, as the therapeutic effects of game therapy have been examined in several studies [26]. For example, researchers have investigated the effectiveness of virtual reality and biofeedback-assisted video games in reducing stress, anxiety, and depression [27, 28]. While the therapeutic effects of these approaches rely largely on the specific activities performed in the game, some studies have pointed out that music plays a crucial role in creating a soothing ambiance and cohering with other game elements [28].

More related to this study, several studies have explored the role of music as an emotional engine in gameplay and have identified its potential therapeutic effects, such as emotional training and the modification of disruptive behaviours in children and adolescents [29, 30]. Furthermore, a few studies have investigated the potential use of rhythm games as a means of treating certain psychiatric conditions, such as helping children with ADHD develop social skills and training speech motor skills in children with speech difficulties [31, 32]. These studies suggest that using music rhythm games for therapy is a practical and promising approach that warrants further exploration. By leveraging the therapeutic potential of music rhythm games, digital active music therapy can provide a novel and engaging therapeutic experience for individuals who may not have access to traditional face-to-face therapy and could expand the range of therapeutic options available to them.

3. EXPERIMENT METHODOLOGY AND SETUP

A. REPRESENTATION OF EMOTIONS

There are two primary methods for representing emotions in music: categorical and dimensional psychometrics. Categorical approaches use discrete emotional descriptors, or tags, to classify musical moods. In contrast, dimensional approaches map musical moods into continuous, multidimensional spaces. This paper adopts the widely-accepted two-dimensional Russell's Valence-Arousal (V-A) model [33] to represent musical emotions. The model's validity and reliability are supported by previous studies [34, 35, 36], and it is widely used in music emotion research, facilitating easy referencing and comparison with other studies. Moreover, the dimensional V-A model allows for a more intuitive observation of the impact of valence and arousal values on listeners' emotions as they change gradually in music. The Russell's V-A model comprises the valence and arousal axes, where low to high arousal values represent a shift from a calm to a more energetic emotional state, and negative to positive valence values indicate a change from a negative to positive emotional state or mood. The model can be further subdivided into four quadrants which represent four main emotion categories: happy, relaxed, sad and angry, as shown in **Figure 1** with other nuanced categories mapped to the plane.

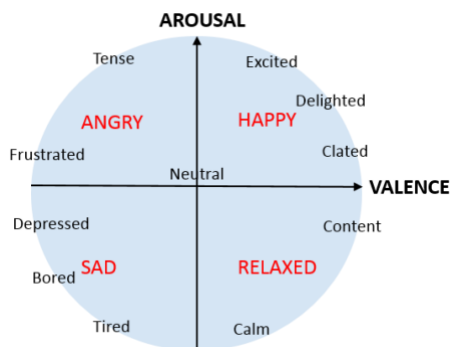


Figure 1: Russell's Valence-Arousal Model [37].

B. MEASUREMENTS OF EMOTIONAL STATUS

I. LONG-TERM ANXIETY AND DEPRESSION

To determine the potential therapeutic effects of digital music rhythm games on anxiety and depression, it is crucial to assess participants' anxiety and depression levels as it allows for the observation of the therapeutic effects for different anxiety and depression groups. In this paper, we use the Patient Health Questionnaire-9 (PHQ-9) and General Anxiety Disorder-7 (GAD-7) questionnaire to briefly assess the level of anxiety and depression in participants. These questionnaires were selected due to their brevity, which allows for more time to be dedicated to music rhythm gameplay during the listening tests. Additionally, their validity has been established in numerous studies [38, 39, 40, 41], providing a brief yet accurate overview of the participants' long-term mental state.

II. CURRENT EMOTIONAL STATE

In order to observe the effect of music therapy via the digital rhythm game format in improving listeners' emotions, it is necessary to assess the emotional state of the listeners both before and after the rhythm gameplay for comparison. In this study, the Self-Assessment Manikin (SAM) scale is used to measure participants' emotional states before and after the test. The SAM, as depicted in **Figure 2**, employs a pictorial representation to assess valence and arousal values on a 9-point scale. This simple and efficient tool offers reliable and valid measurement for affective response [42, 43, 44]. Furthermore, its pictorial nature provides a visual representation that allows participants to intuitively assess their emotions, avoiding the ambiguity of categorical descriptions.

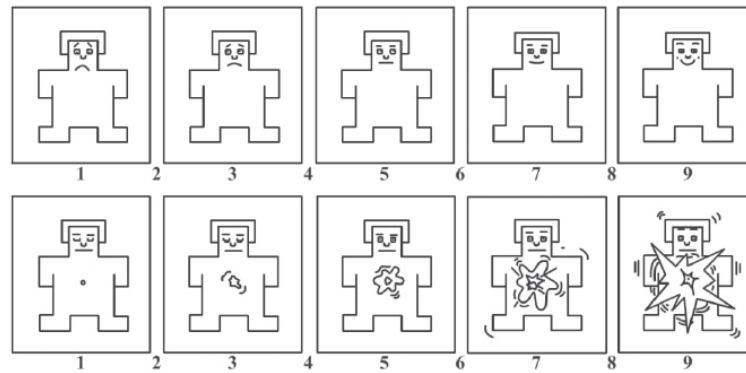


Figure 2: Self-Assessment Manikin (SAM) for valence (upper) and arousal (lower) [45]

C. STIMULI AND PROPOSED METHODS

For the purpose of easy comparison and referencing with our previous study on music therapy [24], we employed the same stimuli dataset. The dataset consists of approximately 4000 songs from Deezer [46] and MIR CISUC [47], with valence and arousal values that have been annotated and verified by experts. To avoid offense, we conducted a manual inspection to remove songs with explicit language that could potentially cause disturbance. Additionally, songs longer than five minutes were excluded to control the overall duration of the listening test to prevent fatigue and deterioration of attention levels. After applying the filtering criteria, a total of 3,137 songs were selected as stimuli for the listening test.

In our previous research, we observed that uplifting and relaxing music, which differ mainly in arousal values in the Russell's V-A model, can effectively enhance listeners' valence or mood. The present study aims to further investigate the impact of uplifting and relaxing music on changing listeners' emotions while utilising the active music therapy approach. To determine the optimal sequencing of songs with varying arousal levels to enhance listeners' emotions, we propose five song sequencing methods: random, all-relaxed, all-upbeat, relaxed-to-upbeat, and upbeat-to-relaxed. **Table 1** presents the details of each method. For each method, a 5-song list is dynamically generated during the listening test from the 3,137 filtered songs with specific ordering according to the respective method. These songs are assigned into four quadrants (happy, relaxed, sad, and angry) based on the V-A model. For the random method, songs will be randomly selected from all four quadrants for baseline comparison with other methods. For the all-relaxed and all-upbeat methods, songs will only be selected from these respective quadrant. For the relaxed-to-upbeat and upbeat-to-relaxed methods, songs within the relaxed and happy quadrants will be sorted by their arousal values and divided into five partitions. One song will be randomly selected from each partition to form the 5-song list, arranged in ascending and descending arousal values for the relaxed-to-upbeat and upbeat-to-relaxed methods, respectively. This approach ensures that selected songs from both quadrants are included, and arousal values are evenly distributed.

Table 1: Proposed Methods for Song Sequencing in Music Rhythm Game.

Method	Description
random	Songs are picked randomly from music dataset
all-upbeat	All songs are picked from happy quadrant
all-relaxed	All songs are picked from relaxed quadrant
relaxed-upbeat	Songs are picked from happy and relaxed quadrant, arranged in increasing arousal
upbeat-relaxed	Songs are picked from happy and relaxed quadrant, arranged in decreasing arousal

D. MUSIC RHYTHM GAMES

The particular focus of our study on active music therapy is on music rhythm games. For the purpose of our investigation, we have selected the music rhythm game *osu!* as the medium for our listening test. This game enjoys immense popularity with over 19.3 million monthly active users worldwide [48] and is ranked 32nd among all PC games [49]. Furthermore, the open-source nature of *osu!* and its thriving user community provides a platform for easy manipulation and customization to facilitate our listening test, which employs the classical Taiko mode offered by *osu!* among its various rhythm game modes. This particular mode is widely recognized as a classic within the genre and is likely to be familiar to a considerable number of participants. Additionally, the gameplay mechanics of the Taiko mode involves only two operations, making it easy to understand and accessible to individuals who may not have experience with music rhythm games. The interface of the *osu!* gameplay is depicted in **Figure 3**.

The playable component, where players can hit notes according to the rhythm, is called beatmaps in *osu!*. To prepare the beatmaps for our stimuli, we first determined the beats per minute (bpm) and time offset for each song by utilising TimeAnalyzer, a tool developed by members of the *osu!* community [50]. Subsequently, we employed an automatic beatmap generator called *osumapper*, also created by *osu!* community members [51], to generate beatmaps for each song. To ensure that the beatmaps were manageable for all participants, we set the overall difficulty parameter to 0. To streamline the process of generating beatmaps for all stimuli, we developed a Python script for batch processing. The Python code utilized in generating the beatmaps for our study can be retrieved at [52].

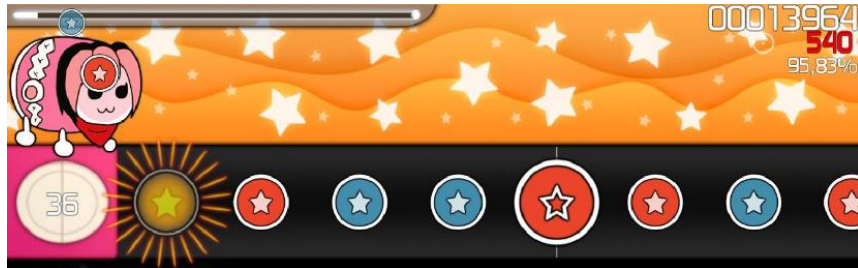


Figure 3: *osu!* Taiko Mode Game Interface and Operations: Left Mouse Click for Red Notes, Right Mouse Click for Blue Notes.

E. LISTENING TEST SETUP

We recruited 89 undergraduate students, with 26 female and 63 male, from the Hong Kong University of Science and Technology in two semesters, Fall 2022 and Spring 2023, to participate in our online listening test. To ensure optimal conditions for the test, we instructed participants to remain in a quiet room and wear earphones to minimise any potential auditory distractions. Additionally, participants were required to record both themselves and their screen activity to confirm their adherence to the instructions and active engagement in the game throughout the listening test. By employing these measures, we aimed to create an ideal testing environment that would facilitate reliable and valid data collection.

The listening test consisted of three primary sections, namely the Pre-test survey, Gameplay, and Post-test survey, and was designed to be completed within a time frame of approximately 30 minutes.

a. Pre-test Survey

At the beginning of the listening test, participants were requested to provide basic personal information for demographic analysis. Subsequently, participants were presented with the GAD-7 and PHQ-9 questionnaires to assess their anxiety and depression level. Following the questionnaires, participants indicated their current valence and arousal states using the Self-Assessment Manikin (SAM) scale.

b. Gameplay

In the study, each participant was randomly assigned one of the five proposed methods. Subsequently, a 5-song list was dynamically generated based on the allocated method and automatically imported into *Osu!*. To prevent any premature expectations or biases based on song titles, the five songs were numbered from “1” to “5” without explicit identification. Participants played through the five songs in accordance with their assigned numerical order.

c. Post-test Survey

Following gameplay, participants were asked to reassess their valence and arousal levels using the SAM scale.

4. RESULTS

A total of 219 samples were collected in listening tests. The total number of participants was 89, with 26 female and 63 male subjects. Twenty-nine of the subjects had some musical background.

One-way repeated measures ANOVA was used to compare the pre- and post- test arousal and valence values reported by the participants through the SAM scale for the five proposed methods (random, all-relaxed, all-upbeat, relaxed-to-upbeat, upbeat-to-relaxed) at the $p < 0.05$ level.

A. OVERALL

The ANOVA test was applied to all 219 collected samples, and the results indicated that all five proposed methods had a significant effect on altering arousal levels in participants, as shown in **Table 2**.

All five methods produced positive changes in arousal levels. Among the methods, all-relaxed resulted in the largest positive change in arousal, closely followed by all-upbeat, and then upbeat-to-relaxed.

For valence, all the methods except all-upbeat produced a significant positive change in the valence as shown in **Table 3**, though the magnitude of the valence change was about half that as for arousal. Upbeat-to-relaxed produced the largest positive change in valence, with relaxed-to-upbeat and all-relaxed about the same level. All-upbeat was much worse than the others.

Taken together, all-relaxed and upbeat-to-relaxed were the most effective methods for increasing both arousal and valence in a strong significant way. The calming effect of all-relaxed music perhaps helped subjects enjoy the gameplay, and upbeat-to-relaxed seemed to help players stay focused and positive even when gameplay become more fast and furious.

Table 2: Overall ANOVA results, ranked by change in arousal for each method

Method	F Value	Num DF	Den DF	Pr > F	Arousal (Pre)	Arousal (Post)	Arousal Change
all-relaxed	22.802	1	47	0.000	3.583	4.750	1.167
all-upbeat	12.295	1	41	0.001	3.619	4.691	1.071
upbeat-to-relaxed	27.666	1	53	0.000	4.463	5.426	0.963
relaxed-to-upbeat	8.298	1	27	0.008	3.821	4.571	0.750
random	4.645	1	46	0.036	3.894	4.426	0.532

Table 3: Overall ANOVA results, ranked by change in valence for each method

Method	F Value	Num DF	Den DF	Pr > F	Valence (Pre)	Valence (Post)	Valence Change
upbeat-to-relaxed	10.005	1	53	0.003	5.204	5.815	0.611
relaxed-to-upbeat	4.917	1	27	0.035	5.393	6.000	0.607
all-relaxed	11.690	1	47	0.001	5.438	6.021	0.583
random	5.659	1	46	0.022	5.128	5.596	0.468
all-upbeat	0.622	1	41	0.435	5.452	5.643	0.191

B. CONSIDERING INITIAL AROUSAL AND VALENCE LEVELS

The initial emotional state of participants may influence the effectiveness of the therapy. Thus, the collected samples were grouped based on their initial levels into Low (1-3), Mid (4-6), and High (7-9) for arousal and Negative (1-3), Balanced (4-6) and Positive (7-9) for valence. ANOVA tests were then applied to investigate the effectiveness of the proposed methods within each group. The complete ANOVA results for all initial Arousal and Valence groups can be found in the **Table 4** and **Table 5** respectively.

For the Low group, all-upbeat and all-relaxed had the largest increase in arousal. For the Mid group, all-relaxed had the largest increase. We excluded the High group since there were usually only 1-2 subjects. In general, the Low group naturally had larger increases compared to the Mid group, since there was more room for improvement. While all-upbeat gave the largest increase for the Low group, it was the least effective for the Mid group. All-relaxed gave the largest increase for the Mid group, and was second best for Low, which explains why it was the best overall in **Table 2**. It is an example of how to use different methods to target different groups: upbeat songs for low-energy individuals and relaxed songs for others to perk up their energy levels.

For valence, the Negative group also had larger increases than Balanced, since there was more room for change. All-upbeat and all-relaxed were best for the Negative group while upbeat-to-relaxed was best for the Balanced group. There were not enough Positive subjects to draw clear conclusions. Once again, while all-upbeat had the largest increase for the Negative group, it was the least effective for the Balanced group. Contrastingly, while upbeat-to-relaxed was the least effective for the Negative group, it gave the largest increase for the Balanced group.

Table 4: ANOVA Results Ranked by Change in Arousal for Different Initial Arousal Levels

Initial Level	Method	F Value	Num DF	Den DF	Pr > F	Arousal (Pre)	Arousal (Post)	Arousal Change
Low	all-upbeat	16.206	1	22	0.001	2.348	4.087	1.739
	all-relaxed	14.248	1	24	0.001	2.520	4.000	1.480
	upbeat-to-relaxed	13.091	1	18	0.002	2.474	3.737	1.263
	random	8.944	1	17	0.008	2.000	3.222	1.222
	relaxed-to-upbeat	1.138	1	11	0.309	2.167	2.667	0.500
Mid	all-relaxed	11.561	1	20	0.003	4.524	5.476	0.952
	relaxed-to-upbeat	8.291	1	13	0.013	4.714	5.643	0.929
	upbeat-to-relaxed	8.827	1	25	0.007	4.962	5.769	0.808
	random	2.281	1	25	0.144	4.769	5.154	0.385
	All-upbeat	0.894	1	16	0.358	4.941	5.294	0.353

Table 5: ANOVA Results Ranked by Change in Valence for Different Initial Valence Levels

Initial Level	Method	F Value	Num DF	Den DF	Pr > F	Valence (Pre)	Valence (Post)	Valence Change
Negative	all-upbeat	40.000	1	4	0.003	2.800	4.800	2.000
	all-relaxed	13.364	1	3	0.035	2.750	4.500	1.750
	relaxed-to-upbeat	1.69E+30	1	1	0.000	2.500	3.500	1.000
	random	1.185	1	4	0.338	2.600	3.400	0.800
	upbeat-to-relaxed	1.412	1	6	0.280	2.714	3.286	0.571

Balanced	upbeat-to-relaxed	17.689	1	36	0.000	5.108	5.973	0.865
	relaxed-to-upbeat	5.444	1	18	0.031	4.947	5.737	0.789
	random	6.478	1	32	0.016	4.909	5.515	0.606
	all-relaxed	2.588	1	32	0.118	5.212	5.545	0.333
	all-upbeat	0.054	1	26	0.819	5.185	5.111	-0.074

C. CONSIDERING GAD-7 AND PHQ-9 GROUPINGS

The long-term anxiety and depression levels of participants may also influence the effectiveness of the therapy. To investigate this, ANOVA tests were applied to different groupings of participants based on their GAD-7 and PHQ-9 scores from the pre-test questionnaire. The GAD-7 scores were categorized as follows: Minimal Anxiety (score 0-4), Mild Anxiety (score 5-9), Moderate Anxiety (score 10-14), and Severe Anxiety (score 15-21). The PHQ-9 scores were categorized as: Minimal Depression (score 0-4), Mild Depression (score 5-9), Moderate Depression (score 10-14), and Severe Depression (score 15-27).

We compared the GAD-7 and PHQ-9 scores and the distribution is shown in **Figure 4** and by categories in **Table 6**. Most of the points are close to the regression line in **Figure 4** or the main diagonal in **Table 6**, with slightly more above the diagonal than below indicating more depression than anxiety among subjects. The GAD-7 and PHQ-9 scores had a strong correlation of 0.823, indicating that the levels of depression and anxiety move mostly together hand-in-hand for our subjects. There were only three outliers in **Figure 4**.

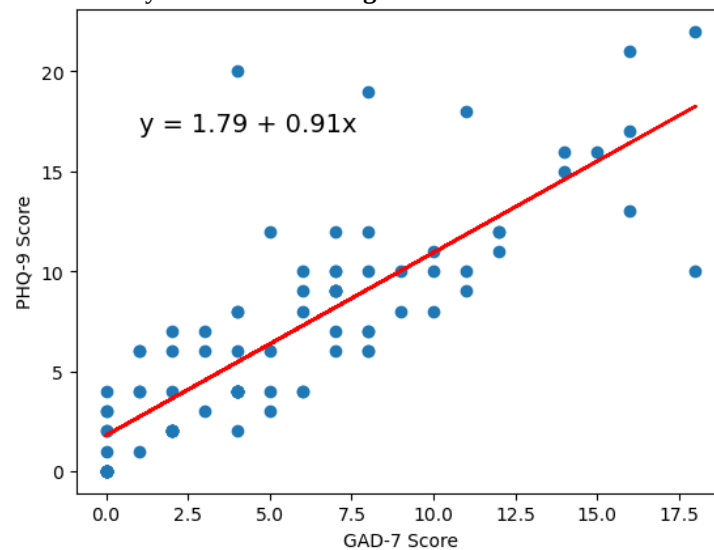


Figure 4: Distribution of GAD-7 and PHQ-9 scores and the corresponding regression line

Table 6: Distribution of GAD-7 and PHQ-9 Scores by Categories

PHQ-9 \ GAD-7	Minimal	Mild	Moderate	Severe
Severe	1	1	3	4
Moderate	0	8	6	2
Mild	9	14	2	0
Minimal	35	4	0	0

The ANOVA results for the GAD-7 groups are shown in **Table 7** and **Table 8** for arousal and valence, respectively. The corresponding PHQ-9 groupings are shown in **Table 9** and **Table 10**. Since there were only a few subjects and the results were not significant, we deleted the Severe group for both the GAD-7 and PHQ-9 tables.

For the GAD-7 arousal results, **Table 7** shows that all-relaxed had the largest increase for the Moderate groups and overall, while upbeat-to-relaxed had the largest increase for Mild. All-upbeat was best for minimal.

For the GAD-7 valence results in **Table 8**, all-relaxed increased the most for Minimal, upbeat-to-relaxed for Mild, and random for Moderate. The valence changes were larger for Moderate subjects than for Minimal and Mild, but the number of subjects was small.

For the PHQ-9 arousal results, **Table 9** shows that all-relaxed increased most for Minimal and Moderate groups, and upbeat-to-relaxed increased most for Mild. There were larger and more significant changes for Mild than Minimal and Moderate.

For the PHQ-9 valence results in **Table 10**, all-relaxed was best for Minimal, upbeat-to-relaxed was best for Mild, and relaxed-to-upbeat was best for Moderate. There were larger and more significant changes for Moderate than Minimal and Mild.

Table 7: ANOVA Results Ranked by Change in Arousal for Different GAD-7 Groupings

GAD-7 Level	Method	F Value	Num DF	Den DF	Pr > F	Arousal (Pre)	Arousal (Post)	Arousal Change
Minimal	all-upbeat	7.719	1	23	0.011	3.125	4.292	1.167
	all-relaxed	10.587	1	26	0.003	3.704	4.852	1.148
	upbeat-to-relaxed	14.271	1	29	0.001	4.400	5.367	0.967
	relaxed-to-upbeat	2.664	1	13	0.127	3.500	4.214	0.714
	random	0.259	1	14	0.619	2.800	3.000	0.200
Mild	upbeat-to-relaxed	20.892	1	12	0.001	4.846	6.154	1.308
	all-relaxed	6.996	1	13	0.020	3.500	4.714	1.214
	relaxed-to-upbeat	9.000	1	9	0.015	4.500	5.500	1.000
	all-upbeat	2.248	1	12	0.160	4.308	5.154	0.846
	random	3.155	1	19	0.092	4.300	5.050	0.750
Moderate	all-relaxed	5.444	1	4	0.080	2.800	4.200	1.400
	upbeat-to-relaxed	3.692	1	6	0.103	4.429	5.571	1.143
	random	2.625	1	6	0.156	4.429	5.429	1.000
	all-upbeat	1.000	1	3	0.391	4.500	5.500	1.000
	relaxed-to-upbeat	1.000	1	2	0.423	3.667	4.333	0.667

Table 8: ANOVA Results Ranked by Change in Valence for Different GAD-7 Groupings

GAD-7 Level	Method	F Value	Num DF	Den DF	Pr > F	Valence (Pre)	Valence (Post)	Valence Change
Minimal	all-relaxed	8.644	1	26	0.007	5.926	6.519	0.593
	relaxed-to-upbeat	1.560	1	13	0.234	6.214	6.643	0.429
	random	2.059	1	14	0.173	6.067	6.400	0.333
	upbeat-to-relaxed	0.683	1	29	0.415	5.933	6.167	0.233
	all-upbeat	0.411	1	23	0.528	6.125	6.292	0.167
Mild	upbeat-to-relaxed	9.561	1	12	0.009	4.692	5.769	1.077
	all-relaxed	2.936	1	13	0.110	4.929	5.429	0.500
	relaxed-to-upbeat	0.546	1	9	0.479	4.600	5.000	0.400
	random	0.192	1	19	0.666	5.100	5.000	-0.100
	all-upbeat	1.361	1	12	0.266	5.000	4.462	-0.538
Moderate	random	32.111	1	6	0.001	4.000	6.429	2.429
	all-upbeat	22.091	1	3	0.018	3.500	5.750	2.250
	relaxed-to-upbeat	12.000	1	2	0.074	5.000	7.000	2.000
	all-relaxed	16.000	1	4	0.016	4.600	6.200	1.600
	upbeat-to-relaxed	3.692	1	6	0.103	4.000	5.143	1.143

Table 9: ANOVA Results Ranked by Change in Arousal for Different PHQ-9 Groupings

PHQ-9 Level	Method	F Value	Num DF	Den DF	Pr > F	Arousal (Pre)	Arousal (Post)	Arousal Change
Minimal	all-relaxed	8.298	1	21	0.009	3.773	5.000	1.227
	all-upbeat	4.077	1	18	0.059	3.053	4.105	1.053
	upbeat-to-relaxed	17.296	1	32	0.000	4.545	5.485	0.939
	random	1.000	1	18	0.331	2.947	3.316	0.368
	relaxed-to-upbeat	0.510	1	10	0.492	3.364	3.727	0.364
Mild	upbeat-to-relaxed	9.315	1	7	0.019	4.625	6.250	1.625
	relaxed-to-upbeat	41.263	1	8	0.000	4.556	6.111	1.556
	all-upbeat	6.336	1	13	0.026	3.857	5.071	1.214
	all-relaxed	8.750	1	14	0.0104	3.467	4.467	1.000
	random	2.161	1	12	0.1673	4.308	5.154	0.846
Moderate	all-relaxed	4.435	1	7	0.073	3.250	4.625	1.375
	all-upbeat	1.724	1	7	0.231	4.375	5.250	0.875
	upbeat-to-relaxed	0.304	1	7	0.598	4.000	4.250	0.250
	random	0.304	1	7	0.5983	4.625	4.875	0.250
	relaxed-to-upbeat	0.125	1	6	0.736	3.857	4.000	0.143

Table 10: ANOVA Results Ranked by Change in Valence for Different PHQ-9 Groupings

PHQ-9 Level	Method	F Value	Num DF	Den DF	Pr > F	Valence (Pre)	Valence (Post)	Valence Change
Minimal	all-relaxed	13.307	1	21	0.002	6.227	6.955	0.727
	upbeat-to-relaxed	2.218	1	32	0.146	5.758	6.152	0.394
	all-upbeat	0.703	1	18	0.413	6.105	6.368	0.263
	random	2.027	1	18	0.172	5.947	6.211	0.263
	relaxed-to-upbeat	0.061	1	10	0.810	6.364	6.455	0.091
Mild	upbeat-to-relaxed	2.272	1	7	0.176	5.000	5.875	0.875
	relaxed-to-upbeat	0.787	1	8	0.401	5.111	5.667	0.556
	all-relaxed	1.556	1	14	0.233	5.000	5.400	0.400
	random	0.040	1	12	0.844	5.077	5.154	0.077
	all-upbeat	0.317	1	13	0.583	5.214	4.929	-0.286
Moderate	relaxed-to-upbeat	12.790	1	6	0.012	4.286	5.571	1.286
	upbeat-to-relaxed	24.652	1	7	0.002	3.375	4.500	1.125
	all-upbeat	3.330	1	7	0.111	4.250	5.125	0.875
	random	2.882	1	7	0.133	4.250	5.125	0.875
	all-relaxed	5.727	1	7	0.048	4.750	5.500	0.750

5. DISCUSSION

A. OVERALL

The findings showed that all methods, including the baseline random method, had significant effects on positively increasing arousal values. In other words, subjects felt more energetic after playing the music rhythm game. This is in contrast to the results of our previous study [24], where with simple passive listening the proposed methods did not show a significant change in arousal. The results in this study indicate that adding engaging elements, such as tapping to the beat of a music rhythm game, boost the effect of music and raise arousal. This may be due to the cognitive demands of matching hand movements to the beat, which requires sustained attention and mental effort, thereby creating a sense of excitement and challenge for players. This, in turn, leads to increased arousal.

The results showed that valence and mood were also significantly increased, though not as much as arousal. The one exception was for the all-upbeat method, where perhaps players found the music too fast for too long and had trouble keeping up, leading to feelings of pressure and frustration that dampened the results. In contrast, when upbeat songs were mixed with slower songs, it allows players to loosen up during the gameplay, leading to a more positive experience and improved mood.

Interestingly, the methods that contained songs of a single type (i.e., all-relaxed and all-upbeat) resulted in a larger increase in arousal, while methods with mixed tempos of music (i.e., relaxed-to-upbeat and upbeat-to-relaxed) resulted

in larger increase in valence. A possible explanation may be that through repeated gameplay, participants became familiar with the rhythm and tempo of the music. As the music continued in a homogenous way, it became easier for them to synchronize their movements with the beat, leading to improved performance and immersion in the gameplay. Consequently, players were more likely to experience a heightened state of arousal. On the other hand, the superiority of mixed-tempo music at improving valence may be attributed to the hedonic adaptation, a diminishing response to stimuli that are affectively relevant due to repeated exposure [53]. Continuous exposure to a single tempo may become boring after a while. Conversely, a sequence of mixed-tempo music perhaps helped to keep it fresh and novel throughout the gameplay, enhancing mood in a positive way.

B. CONSIDERING INITIAL AROUSAL AND VALENCE LEVELS

We wanted to see if there were differences in response depending on participants' initial arousal and valence levels. The findings suggest that participants with lower initial arousal values had more improvement in arousal. The results also suggest ways to customize the systems for different types of subjects. For Low-arousal and Negative-valence subjects, all-upbeat and all-relaxed are both strong for improving arousal and valence alike. For Mid-arousal and Balanced-valence subjects, upbeat-to-relaxed is best.

C. CONSIDERING GAD-7 AND PHQ-9 GROUPINGS

An interesting pattern emerged from the results when grouping subjects by different levels of anxiety and depression: All-relaxed was always the best (or near-best) for subjects in the Minimal group, and upbeat-to-relaxed was always best for subjects in the Mild group. Surprisingly, these results hold for both anxiety and depression. For Moderate subjects, all-relaxed gave the best results for arousal, and gave good improvements (at least +0.75) for valence.

6. CONCLUSION AND FUTURE DIRECTIONS

A. CONCLUSION

This study has investigated a novel approach to active music therapy through a music rhythm games with Hong Kong university student subjects. The results indicate that all proposed song sequencing methods, including random, all-relaxed, all-upbeat, upbeat-to-relaxed, and relaxed-to-upbeat, can positively increased arousal levels, while nearly all methods significantly increased valence levels.

Our findings suggest that all-relaxed and all-upbeat gave the best increase in arousal (i.e., energy level), and that mixed-tempo methods (e.g. upbeat-to-relaxed and relaxed-to-upbeat) gave the best increase in valence (i.e., mood). All-relaxed and upbeat-to-relaxed were most effective in increasing both arousal and valence.

We also grouped subjects by different initial valence and arousal levels, which allows nuanced recommendations for subjects when their initial mood and energy level are known. Similarly, we also found strong enough data for subjects with minimal, mild and moderate levels of anxiety and depression to be able to make tailored recommendations for subjects when their anxiety and depression levels are known, at least for university student subjects like ours when anxiety and depression levels were strongly correlated.

These results highlight the effectiveness of music rhythm games as a tool for improving mood and energy levels in a therapeutic way. Given that music rhythm games are already a popular entertainment activity for youngsters, our results show how to use them to improve mood in a lightweight, convenient, entertaining, and easily accessible self-administered way that does not require supervision or musical background, unlike traditional music therapy.

B. FUTURE DIRECTIONS

This study has mainly focused on relaxing and uplifting music, which we found to be effective in improving valence in passive listening in our previous study. Future research can investigate other music categories, such as heartbreak songs and emotion-releasing angst music and compare their effectiveness.

Additionally, expanding the study to a wider audience will provide a more comprehensive understanding. There are three potential directions for expanding the sample size in our study. Firstly, we can include a greater number of severe cases in both anxiety and depression, as the current study mainly focused on the minimum, mild, and moderate groups due to the limited number of subjects in the severe group. This inclusion would provide a more comprehensive comparison and understanding of the psychological impact across different severity levels.

Secondly, since the number of subjects without a music background outweighed those with a music background, it was challenging to make a fair comparison between the two groups. To address this, future research can strive to include a more balanced representation of subjects with and without a music background. This approach would enable further investigation on the impact of music background on anxiety and depression.

Thirdly, our study revealed an intriguing pattern among subjects from the Hong Kong University of Science and Technology, wherein anxiety and depression exhibited a linear relationship, with slightly more depression than anxiety. To determine whether this linear pattern is unique to HKUST or a general trend among Hong Kong university

students, it would be beneficial to recruit subjects from other universities in Hong Kong. This broader sample would provide insights into the psychological characteristics of Hong Kong university students, enabling the customization of mental health care support for this particular group.

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